

太赫兹通感一体化系统的动态波束跟踪与功率分配

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摘要: 第六代移动通信 (6G) 网络将实现从「万物互联」到「万物智联」的跨越, 除了传统通信功能外, 还将集成高精度感知能力, 支撑自动驾驶、数字孪生、工业物联网等新兴应用场景。通感一体化 (Integrated Sensing and Communication, ISAC) 作为 6G 的核心技术之一, 通过共享硬件平台、无线资源与信号处理模块, 可显著提升频谱效率与系统效能, 降低硬件成本。太赫兹 (Terahertz, THz) 频段 (0.1 10 THz) 拥有海量可用带宽, 可支持 100 Gbps 以上的超高通信速率, 同时其短波长特性可实现厘米级波束宽度, 为高精度感知提供了物理基础, 因此太赫兹通感一体化 (THz-ISAC) 成为近年来的研究热点。然而, THz-ISAC 系统的实用化仍面临诸多挑战, 本文提出了一种基于双深度 Q 网络 (Double Deep Q-Network, DDQN) 的 THz-ISAC 动态波束跟踪与功率分配联合优化算法, 通过将优化问题建模为马尔可夫决策过程, 设计了适配的状态与动作空间, 采用 DDQN 算法实现动态环境下的联合优化。仿真结果表明, 所提算法相比传统算法显著降低了波束失配率, 提升了通感总效用, 且能够适应高速移动场景。

关键词: 太赫兹; 通感一体化; 波束跟踪; 功率分配; 深度强化学习

Dynamic Beam Tracking and Power Allocation for Terahertz Integrated Sensing and Communication Systems

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Abstract: The sixth-generation (6G) mobile communication network will realize the leap from "Internet of Everything" to "Intelligent Internet of Everything". In addition to traditional communication functions, it will also integrate high-precision sensing capabilities to support emerging application scenarios such as autonomous driving, digital twin, and industrial Internet of Things. Integrated Sensing and Communication (ISAC), as one of the core technologies of 6G, can significantly improve spectrum efficiency and system effectiveness, and reduce hardware costs by sharing hardware platforms, wireless resources, and signal processing modules. The terahertz (THz) frequency band (0.1 10 THz) has massive available bandwidth, which can support ultra-high communication rates above 100 Gbps. At the same time, its short wavelength characteristics can achieve centimeter-level beam width, providing a physical basis for high-precision sensing. Therefore, terahertz integrated sensing and communication (THz-ISAC) has become a research hotspot in recent years. However, the practical application of THz-ISAC systems still faces many challenges. This paper proposes a double deep Q-network (DDQN) based joint optimization algorithm for dynamic beam tracking and power allocation in THz-ISAC systems. By modeling the optimization problem as a Markov decision process, designing an appropriate state and action space, and adopting the DDQN algorithm to realize joint optimization in dynamic environments. Simulation results show that the proposed algorithm significantly reduces the beam mismatch rate, improves the total utility of sensing and communication, and can adapt to high-speed mobile scenarios compared with traditional algorithms.

Keywords: terahertz; integrated sensing and communication; beam tracking; power allocation; deep reinforcement learning

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